

HUDSON C. SMITH

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EXPERIENCE

GrayMatter Robotics

Los Angeles, CA

Senior Robotics Engineer: Government Projects

Aug 2026 – Present

- Customer-facing technical lead for government-funded research and development (R&D) programs ranging from \$300K-6M+ in robotic grinding, ultrasonic inspection, and cold spray repair
- Source opportunities, write proposals, and scope requirements directly with customers and government stakeholders to devise robotic systems that improve the efficacy of manufacturing and/or repair processes and meet government needs
- Lead hands-on R&D projects building custom python/ROS 2 software and hardware, coordinating with multiple internal teams and leadership stakeholders and managing external relationships through the project execution phase
- Mentor a team of 2 interns and train new hires and interns on the fundamentals of GrayMatter's systems and software stack
- Serve as the primary interviewer for technical screenings of all new candidates for the government projects team
- Serve as a company delegate to the American Foundry Society (AFS); build relationships with defense and manufacturing decision-makers and advance adoption of robotic automation

Robotics Engineer II: Government Projects

Jul 2025 – Aug 2026

- Was responsible for complex system integration: hardware and software development and debugging, system setup, testing and networking, rapid prototyping, and internal coordination with software and mechatronics teams
- Executed internal R&D efforts alongside a robotics software engineer for a new ultrasonic inspection product, devising system requirements for inspection data collection and building custom ROS2 nodes for ultrasonic sensor communication
- Authored proposals for \$4M+ contract value over 5+ projects, awarded \$2M+ with all projects won advancing to deployment
- Supported technical scoping for new commercial and government R&D opportunities, translating customer requirements into concrete concepts, budgets and execution plans
- Delivered final R&D results as demonstrations, technical reports, pilot programs, and deployment-grade systems

Robotics Engineer: Systems & Applications

Apr 2024 – Jul 2025

- Activated a sanding robot for a leading commercial aerospace maintenance and repair center as the primary on-site engineer
- Led on-site commissioning and live troubleshooting of cement truck refurbishment robotic system through customer handoff
- Built robust, production-scale robotic systems: selected scanning poses and parameters, tuned color perception and segmentation models, defined inputs for the trajectory planner, and created a robust front-end user experience
- Developed custom hand-off packages for customers including system maintenance and safety documentation, standard operating procedures, and on-site training of operations staff
- Led initial development of scan-to-path abrasive cutting and belt grinding capabilities with tooling (PushCorp) and abrasive (3M) partners to devise software requirements and determine physics-based process parameters
- Conducted technical site assessments at prospective customer facilities to evaluate automation feasibility for new projects

ABB Corporate Research

Ladenburg, Germany

Robotics & Motion Technologies Researcher

May 2023 – Nov 2023

- Designed and executed controlled stopping experiments on an ABB IRB-1100 robot across 10,000+ unique initial joint positions and various starting velocities, collecting over 30,000 unique data entries for robot stopping trajectories
- Applied machine learning to predict safe stopping distances for industrial robots under ISO/TS 15066 and ISO 10218 using PyTorch to build probabilistic AI safety models from empirical data, achieving root mean square error of less than 1 degree for stopping distances and less than 15ms for stopping times when interpolating within the sampled position/velocity space
- Published findings at VDI Mechatroniktagung Dresden 2024: *Prediction of Stopping Behavior for Robot Safety*

GreenbergFarrow Architecture

New York, NY

Mechanical Engineering Designer II

Dec 2023 – Apr 2024

- Designed mechanical, electrical and plumbing (MEP) systems for Murphy Oil retail sites nationwide in Autodesk Revit
- Coordinated designs with customers as well as architecture, civil, structural, fire protection and construction stakeholders
- Performed coordination calls and site visits as necessary to ensure compliance with design standards and building codes

Mechanical Engineering Intern

Dec 2023 – Apr 2024

- Supported MEP system design and documentation in Autodesk Revit and BIM360 across commercial and retail projects for customers such as Valvoline (national), Chipotle (national), Murphy Oil (national) and Kitchen United (New York City)

EDUCATION

Technische Universität Darmstadt - *M.Sc. Mechanical & Process Engineering*

Darmstadt, Germany | 2022 – 2023

Virginia Tech - *M.S. Mechanical Engineering*

Blacksburg, Virginia | 2021 – 2023

Hamilton College - *B.A. Physics & German Studies*

Clinton, New York | 2017 – 2021

RELEVANT PROJECTS

Robotic Plate Grinding & Ultrasonic Inspection – Steel Rolling Mill | R&D Project Phase I

- Technical lead through R&D execution: integrated industrial robot, 3D vision, ultrasonic sensor and force-controlled tooling; managed government & stakeholder relationship; invited by funding agency to bid for Phase II & III simultaneously

Aerospace MRO Robotic Automation - Commercial Airline | Deployed Robot

- Lead on-site engineer who took the system from R&D into production: devised scanning and process parameters, tuned path planning, wrote SOPs, trained all customer personnel; system remains live under active sustainment contract

Scan-to-Path Abrasive Cutting Pipeline - Leading Global Producer of Industrial Machinery | R&D Demonstration

- Performed on-site technical discovery with a cutting tooling partner, defined software requirements, co-developed software with engineering team for working customer demonstration; capability served as proof point for subsequent contract award

Safe Stopping Distance Models - ABB Corporate Research | Published Author

- Probabilistic ML models for robot stopping behavior under ISO 10218/TS 15066 for dual-channel, safety-rated stopping distance and time modeling; experimentally validated on ABB robotic system and published at VDI Mechatroniktagung
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TECHNICAL SKILLS

Programming: Python (Pandas, PyTorch, NumPy, OpenCV, Matplotlib), ROS 2, ROS 1, MATLAB, C++, C, VBA

Robotics: Field deployment/commissioning, robot programming (ABB, FANUC), robotic grinding & surface finishing, scan-to-path automation, force-controlled manipulation, closed-loop control logic, high-mix and MRO system design, end-of-arm tooling design

Vision & Perception: 2D + 3D scanning, point cloud processing (segmentation), machine vision integration, object classification

ML / AI: Supervised learning, regression & classification, experimental design, data cleaning and analysis

Engineering Tools: NVIDIA Isaac Sim, MeshLab, Autodesk Revit, AutoCAD, SolidWorks, ABB RobotStudio, FANUC RoboGuide

HONORS & PUBLICATIONS

- Deutschlandstipendium Scholarship - TU Darmstadt (merit-based, German Federal Government)
- Fulbright ETA Fellowship - Alternate, Germany
- Nottingham Leadership Scholarship - Delta Phi Chi Chapter, Hamilton College (merit-based)
- Graduate Teaching Assistant - Virginia Tech Physics Department
- Publication: Smith, “Prediction of Stopping Behavior for Robot Safety,” (First Author, VDI Mechatroniktagung, 2023)
- Publication: “Quantifying Squaraine Phase Separation, Aggregate Populations, and Solar Cell Performance with a Correlated Multimethod Approach, (Journal of Physical Chemistry C, 2022)